Supplementary information for

Wind-Aware, Time-Scale Adaptive $PM_{2.5}$ Spatiotemporal Forecasting with Spatial Regularization

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Table S1. Parameter Explanations for the $PM_{2.5}$ Prediction Model. This table lists the input parameters used in the $PM_{2.5}$ concentration prediction model and provides a brief explanation of their impact on air quality.

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Fig.S1. Distribution of Stations in the Beijing-Tianjin-Hebei Region

Fig.S2. Contour maps comparing $PM_{2.5}$ forecasts at different lead times with ground-truth observations. Results are shown for 1h, 6h, 12h, and 24h forecasts across cities in the BTH region under varying pollution levels.

4.6 Sensitivity analysis of input factors

Table S1. Parameter Explanations for PM2.5 Prediction Model

Parameter	Explanation
AQI (Air Quality Index)	A single numerical indicator that integrates multiple pollutant concentrations (e.g., $PM_{2.5}$, PM_{10}, O_3) to represent overall air quality. It helps track general pollution levels and provides indirect insights into $PM_{2.5}$ trends.
CO (Carbon Monoxide)	Concentration of Carbon Monoxide, mainly from vehicle exhaust and combustion sources. CO often co-occurs with $PM_{2.5}$ in urban areas, indicating shared emission origins.
NO_2 (Nitrogen Dioxide)	A reactive nitrogen gas produced mainly by fossil fuel combustion. It is a precursor to secondary particulate formation and correlates with traffic-related $PM_{2.5}$ emissions.
O_3 (Ozone)	A secondary pollutant formed through photochemical reactions involving NO_x and VOCs. Ozone levels influence the oxidative environment and the formation of secondary $PM_{2.5}$.
PM_{10} (Particulate Matter 10)	Particulate matter with diameter ≤10μm, which includes $PM_{2.5}$. PM10 trends often move with PM2.5 and can indicate broader particle pollution.
$PM_{2.5}$（Particulate Matter 2.5）	Fine particulate matter with diameter ≤2.5μm. It poses significant health risks and is the core target variable in the prediction model.
SO_2 (Sulfur Dioxide)	A gaseous pollutant from industrial processes and coal combustion. SO_2 can react in the atmosphere to form sulfate particles, contributing to $PM_{2.5}$ levels.
t2m (2-meter Temperature)	Near-surface air temperature at 2 meters. It influences atmospheric mixing, stability, and chemical reaction rates, all of which affect $PM_{2.5}$ dynamics.
d2m (2-meter Dew Point Temperature)	Dew point temperature at 2 meters, reflecting ambient humidity. High dew points are associated with enhanced aerosol hygroscopic growth and secondary PM formation.

V10 (10-meter North-South Wind Component)	Meridional (north-south) wind component at 10 meters altitude. It affects pollutant transport and regional dispersion of $PM_{2.5}$.
u10 (10-meter East-West Wind Component)	Zonal (east-west) wind component at 10 meters altitude. Together with V10, it determines wind-driven horizontal movement of air pollutants.
tp (Total Precipitation Rate)	Total precipitation rate in millimeters per hour. Rainfall helps remove $PM_{2.5}$ from the air through wet deposition.
sp (Surface Pressure)	Surface atmospheric pressure, indicating synoptic weather conditions. It influences boundary layer height and the vertical mixing of pollutants, thus affecting $PM_{2.5}$ accumulation.

Table S2. Optimal parameter values used in the experiments.

Hyperparameters	Values
Training epoch	100
Input sequence length	168
Hidden dimension	64
Mid-layer channel	32
Number of heads	4
Window size	8
Distance-decay temperature	6.0
Wind-bias scale	0.5
Time-decay temperature	12.0
Number of experts	4
Top-U temperature	0.5
Entropy-reg weight	0.01
Number of prototypes	50
Cluster-softmax temperature	1.0
Dynamic-adj temperature	0.5
Static–dynamic fusion α init	0.1
Laplacian-reg weight	0.1
Learning rate	1e-4
Weight decay	1e-3

Table S3. Air Quality Level Classification Based on the "Technical Regulations for Ambient Air Quality Index"

Level	PM2.5 concentration range (μg/m³)
Excellent	0-35
Good	36-75
Light Pollution	76-115
Moderate Pollution	116-150
Heavy Pollution	151-250
Severe Pollution	>250

Fig.S1. Distribution of Stations in the Beijing-Tianjin-Hebei Region

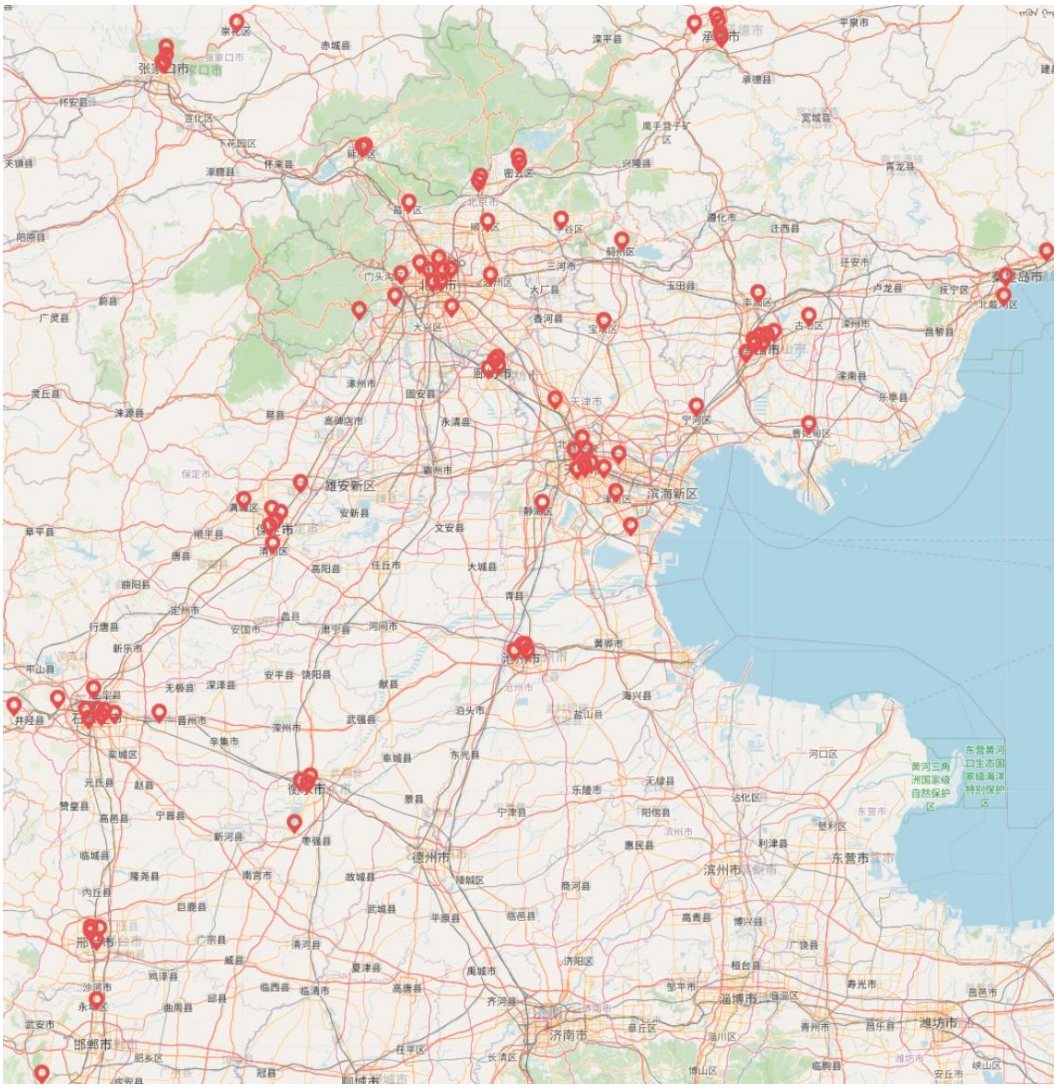
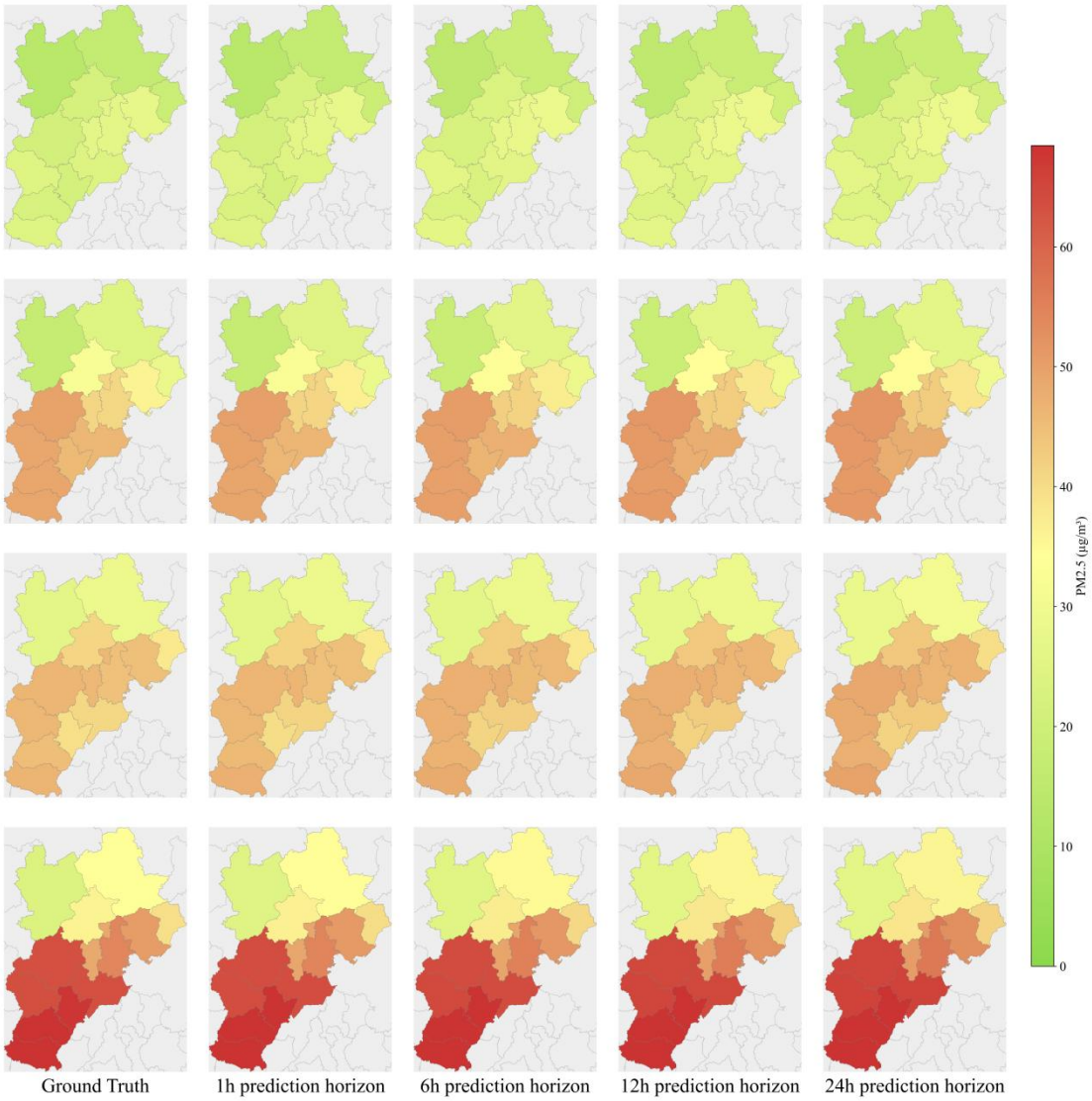


Fig.S2. Contour maps comparing $PM_{2.5}$ forecasts at different lead times with ground-truth observations. Results are shown for 1h, 6h, 12h, and 24h forecasts across cities in the BTH region under varying pollution levels.



4.6 Sensitivity analysis of input factors

To systematically quantify the marginal contribution of different input feature types to$PM_{2.5}$forecasting, we designed four ablation experiments under an identical network architecture and hyperparameter configuration. Each variant removes one category of features: basic meteorological variables, other pollutant information, precipitation, or wind. All ablations follow the baseline training and evaluation pipeline with identical architecture and hyperparameters.The only difference is the input feature subset. The configurations are as follows:

(1) Baseline: retain all 13 input features (AQI,PM10,PM2.5, CO, NO2, SO2, O3, T2M, D2M, SP, TP, U10, V10), asthe performance reference.

(2) w/o climate: remove stability-related meteorological variables (T2M, D2M, SP) to assess the role of the basicmeteorological state.

(3) w/o pollution: keep only the target variablePM2.5along with all meteorological variables (T2M, D2M, SP, TP,U10, V10), removing the other six pollutant variables (AQI,PM_{10}, CO, NO_2, SO_2, O_3).

(4) w/o rain: remove TP while retaining pollutant variables and the remaining meteorological variables (T2M, D2M,SP, U10, V10) to quantify the importance of precipitation.

(5) w/o wind: remove U10 and V10 while retaining pollutant variables and the other meteorological variables (T2M,D2M, SP, TP) to quantify the contribution of wind speed and direction.

For each experiment, we compute MAE, RMSE, and Corr at 1 h, 6 h, 12 h, and 24 h lead times to quantify themarginal contribution of each input feature category.